

Learning Hawaiian Open Ocean Navigation Methods with Kilo Hōkū VR

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ABSTRACT

Kilo Hōkū VR (t: “to observe and study the stars”) is a virtual reality simulation of the Hōkūle‘a, a Polynesian double-hulled sailing canoe, and a simulation of the practice of Modern Hawaiian wayfinding, or non-instrument open ocean navigation. It was developed to assist in the cultural preservation of the celestial navigation portion of Modern Hawaiian wayfinding, and to expand the availability of learning opportunities. We here introduce new features added to the simulation for teacher and student interaction and learning. We performed a user study and observed the potential viability of using Kilo Hōkū VR with students who are currently learning wayfinding in a classroom setting.

Keywords: Virtual Reality, Cultural Heritage, Interactive Learning Environment, Immersion.

Index Terms: Applied computing~Interactive learning environments, Human-centered computing~Virtual reality

1 INTRODUCTION

The Hōkūle‘a is a double-hulled Hawaiian voyaging canoe, also known as a wa‘a kaulua. She was constructed in 1975 with the intent of proving that native Hawaiians journeyed between Tahiti and Hawai‘i with intent and purpose. In 1976, a crew successfully sailed Hōkūle‘a from Hawai‘i to Tahiti using traditional wayfinding methods, and no modern navigational instruments [1].

With the intent of education and preservation of Modern Hawaiian wayfinding, the first release of Kilo Hōkū VR allowed users to sail the Hōkūle‘a within a virtual environment. This allows greater access to the otherwise limited learning opportunities on board a wa‘a kaulua. This initial prototype was completed in late 2017 [2]. After determining future work through our consultation with active practitioners of Modern Hawaiian wayfinding, we set out to improve the user experience and expand the features offered within the simulation, and perform a user study.

2 METHODS

2.1 Development & New Features

The simulation was developed using Unity 3D and run using a physically wired HTC Vive Pro virtual reality headset, and the HTC Vive Pro controller wands. To improve the simulation's usefulness as a learning tool, we gathered feedback from our first prototype

and used it to update the simulation with new visual guides and learning aides to highlight methods used in Modern Hawaiian wayfinding, and tools for teachers to manipulate the state of the simulation. This includes indicators of the celestial meridian and equators, user-friendly contextual button guides that appear around the controllers, teleportation movement, and railing markers for indicators of angles relative to the sailing vessel.

Changes have also been made to the controller wand to functionality mappings. We found that most users in early trials with the prototype simulation had difficulty using certain buttons on the HTC Vive Pro controller wands, namely the side grip buttons and the menu button above the circular touchpad. Functionality was moved to other buttons to compensate for this difficulty.

2.2 User Study

This study was performed with approval of the Institutional Review Board of the University of Hawai‘i at Mānoa, Protocol ID 2018-00840. Guidelines for ethical treatment of human subjects were followed accordingly.

2.2.1 Participants

Participants were a combined 44 students and community members (19 women, 25 men, ages 18-68, $M = 31.83$, $SD = 14.80$). Participants were enrolled in Hawaiian Studies 281 (HWST 281) in the University of Hawai‘i system, or community members who have knowledge of the practices of Modern Hawaiian wayfinding.

2.2.2 Pre-Survey, Simulation Tasks, & Post-Survey

In the pre-survey, participants were asked their age, sex, year in school, whether they had used Virtual Reality in the past, their experience and exposure to Hawaiian sailing canoes, their experience with Celestial Navigation and Modern Hawaiian wayfinding, and what was conceptually difficult for them in learning Celestial Navigation. Participants self-reported on this knowledge.

The simulation was presented to participants via two Dell Alienware laptops, and two HTC Vive Pro virtual reality systems. Participants were presented with separate, isolated versions of the simulation and did not interact with each other within the simulation.

Participants were guided through a series of tasks within the simulation, followed by free time wherein they were allowed to explore the simulation, for a total of 10 minutes. These tasks consisted of being guided through VR basics and movement, usage of certain functionality within the simulation, and performing certain tasks within the simulation related to Modern Hawaiian wayfinding and non-instrument navigation.

After performing tasks and freely exploring the simulation, participants filled out a post survey comprised of 16 questions wherein the participant would make self-ratings on a 6 point scale

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with questions derived from Parong & Mayer's study [3], followed by open response questions about the participant's thoughts on the simulation.

3 RESULTS

3.1 Pre-simulation Survey Questions

For the pre-simulation survey responses, participants responded that most had not previously used some type of virtual reality headset, most had been on board of one of the wa'a kaulua but had not sailed off shore on one, and all participants had familiarity with the Hawaiian star compass.

When responding to what concepts or practices for celestial navigation were difficult to understand, participants noted they had difficulty with memorization of the different stars and star lines, the rising and setting points for stars relative to the star compass, learning the math for determining the declination (angle from the horizon) for a particular star, and Hawaiian names for stars and constellations. Some participants who were students noted that they had not put as much effort into their examination of the topic material as they should have—this would draw interesting parallels later in the observational portion of the study.

3.2 Simulation Observations

During use of the simulation, it was apparent which participants had a genuine interest and engagement with the subject matter. These participants responded with immediate engagement with the simulation by actively attempting to perform the tasks as directed, and actively using the simulation during the free time after the guided instructions. Those participants who had noted that they did not put significant effort into learning the course material were observed as usually standing still and not engaging heavily with the simulation during the free time after the guided instructions.

Nearly all participants were successful in following the provided instructions. The major points of difficulty for participants were finding the star Hōkūpa'a (the North Star), the subsequent question of being able to determine the angular height of the North Star based on other guides in the simulation, and being able to successfully utilize the steering of the vessel.

3.3 Post-simulation Questions

Post-simulation survey responses were mostly positive in nature. Participants self-reported that the subject matter for the study was either not difficult for them, or of middling difficulty. Most students also reported a middling or better understanding of the material in the study, with the lower understanding scores reflected by the observation that some students had difficulty finding the North Star. Cognitive load was reported mostly around the middle range, with slightly heavier weight given towards using less mental effort in the simulation. The majority of participants reported that they liked learning in this manner, would like to do so in the future, and would like to learn about the subject material by using the simulation. This could be due to the nature in which some participants reported that the simulation allowed them "to actually feel as if [they] were on Hokule'a" and allowed them to "[help] relate what we learn conceptually to actual navigation." Because of placement in context and immersion, engagement may be improved as noted by Kampling [5]. This is a potential driving factor for the high motivation during use of the simulation. Engagement was self-reported as generally high by all participants, with reporting that they found the lesson and material useful to them. Again, those with middling to low engagement could be correlated to those who either self-reported as not being as engaged with the material, or who experienced discomfort while using the simulation.

4 DISCUSSION AND FUTURE WORK

The study results were overall positive, with self-reporting from study participants that the simulation put the material in context, and made it easier to visualize and study. However, while this study was built to measure the reactions of students and practitioners of Modern Hawaiian wayfinding to the simulation, it was not built to compare whether learning using the simulation would produce better learning outcomes than the existing instructional methods.

Future work is needed to determine whether learning Modern Hawaiian wayfinding with the addition of the simulation to existing instructional methods would produce desired learning outcomes and assist in idea building and knowledge retention. Furthermore, it cannot be determined from this study whether students would actually benefit from use of the simulation during classroom instruction, only that they perceived benefits from simulation use.

5 CONCLUSION

The main contribution of this work is a template for constructing learner- and practitioner- acceptable wayfinding simulations that can be used to implement other wayfinding simulations in virtual reality for different cultural and non-cultural non-instrument navigation practices. It is also a first-of-its kind simulation of a Hawaiian double-hulled sailing canoe and the practice of Modern Hawaiian wayfinding in virtual reality, and demonstrates that such a simulation is of perceived use for practitioners and students of Modern Hawaiian wayfinding. It is the only virtual reality simulation currently available that covers Polynesian voyaging within context on a double-hulled sailing vessel, and serves as a cultural heritage preservation piece in our modern technological age. It also makes a unique cultural practice available to anyone with the equipment to run the simulation, at low cost and effort to set up and utilize. It adds another tool with advantages over the existing educational methods used for teaching Modern Hawaiian wayfinding in a classroom setting.

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